

601.445/645

Practical Cryptographic Systems

Symmetric Cryptography

Instructor: Matthew Green

Housekeeping

- **Assignment 1 out, right after class**
 - It's on the website, but the link 404s (that's on purpose)
 - I'll make it public after class
- I made some changes to the website
 - Reading is up
 - Removed “notes” stuff

MTU Vigenère cryptanalysis tutorial (esp. frequency analysis, Kasiski, and index-of-coincidence pages); optional: [Friedman, *The Index of Coincidence and Its Applications in Cryptanalysis* \(1922\)](#); background: [book ch. 2 Background \(PDF\)](#) — classical ciphers, the one-time pad, and the Enigma

A1

- **This assignment is a first for me**
 - You can use coding agents (but you don't have to)
 - I've even given you CLAUDE.md and AGENTS.md to help you
 - A lot of your points come from an in-person assessment and test/quiz questions
 - So the point of the assignment is to produce a person who can pass those things, not to produce code

A1

- **I'm also assessing the AI agents**
 - At the end of your project, the agent is instructed to output its own report on how well this worked
 - This report is designed to be non-user-identifying
(you should read it to be sure)
 - I got excited about this yesterday and designed a Tor-based dropbox system that uses ring signatures to allow you to anonymously submit this without identifying yourself

A1

- **Timeline:**
 - **Part 0: Due September 11 EOD,**
I will need you to post a public key to Gradescope
(this is a 5-10 minute process) and you'll need to keep the secret part!
 - **Remaining parts: Due September 18th EOD**
 - **Review signup: the following week**
 - I've added the Gradescope to the pinned Piazza post

News

NEWS / TECHNOLOGY

Apple Takes Its iCloud Encryption Fight to a UK Tribunal



Nate Gregory

Published Aug 3, 2026



News

Russian Hackers Are Silently Reading Your WhatsApp Messages Right Now

By Mauven MacLeod · 10 March 2026

♡ 21

Hello, Mauven here.

Yesterday, the Netherlands' two principal intelligence agencies issued a formal public warning. The AIVD, their domestic security service, and the MIVD, their military intelligence counterpart, confirmed that Russian state-sponsored hackers are running what they describe as a large-scale global campaign to compromise Signal and WhatsApp accounts.

Targets include government officials, military personnel, diplomats, and journalists. Dutch government employees have already been affected. Sensitive information has likely already been accessed.

I want to explain precisely why this matters to your small business.

The Attack Does Not Break the Encryption

This is the most important thing to understand first.

Signal and WhatsApp have robust [end-to-end encryption](#). That encryption has not been broken. The apps themselves have not been compromised. Russian hackers did not find a backdoor into the platform.

News

Method Two: The Silent Linked Device

This one is considerably more insidious, because the victim never loses access to their account.

Both Signal and WhatsApp support a legitimate feature allowing users to link their account to multiple devices: a computer, a tablet, a second phone. On WhatsApp, this is called Linked Devices. The process normally requires scanning a QR code.

Attackers send targets a QR code or link disguised as a group invitation, a security verification prompt, or a resource from the app’s official website. When the target scans it, they unknowingly link the attacker’s device to their account.

The target continues using WhatsApp normally. Their old messages are still there. Nothing appears to have changed. Meanwhile, the attacker’s device is silently receiving every new message sent to or from that account, in real time, potentially indefinitely.

Germany’s BSI advisory notes that Signal lists all linked devices under Settings, but that users rarely check this. That is a reasonable observation: most people have never opened that menu in their lives.

(Finish Monday's slides)

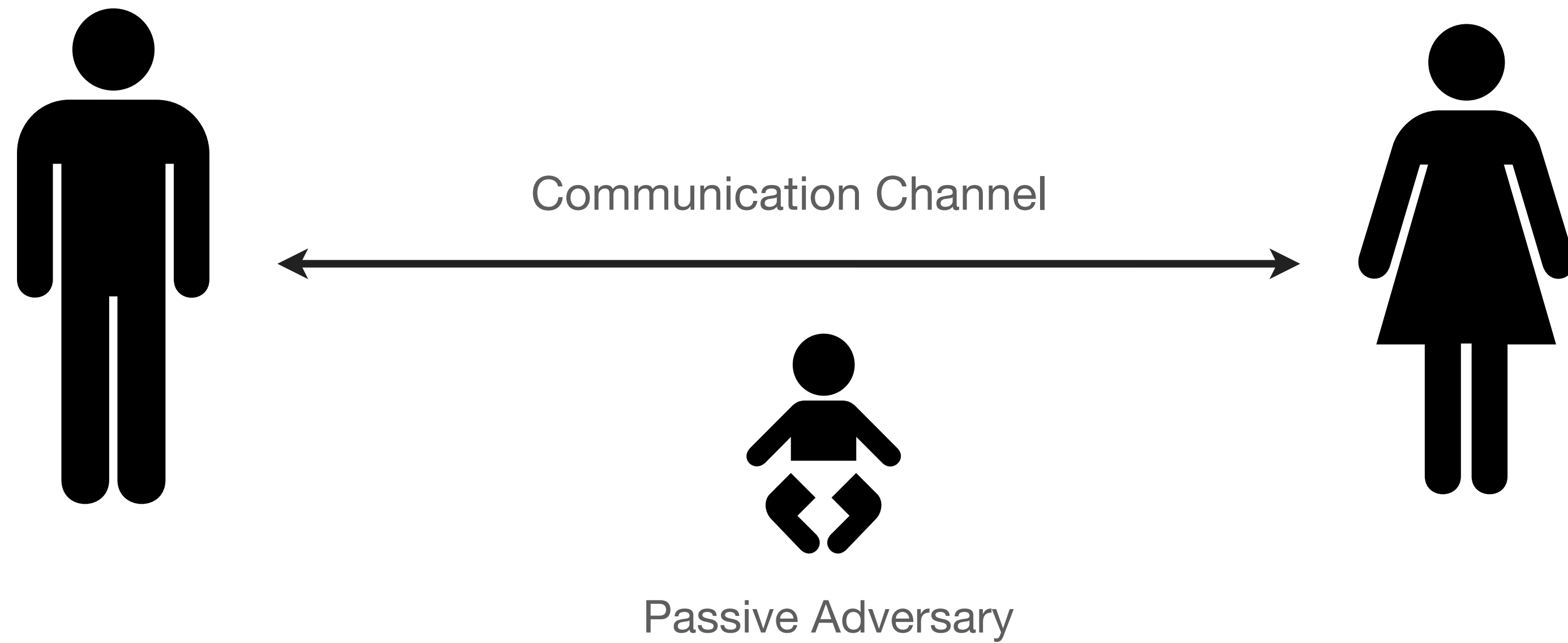
Review

- Last time:
 - A few examples of how systems break
 - Bad primitives, bad protocols, bad implementation
- Today & Weds:
 - A (brief) tour through cryptologic history
 - Starting with symmetric (secret-key) crypto

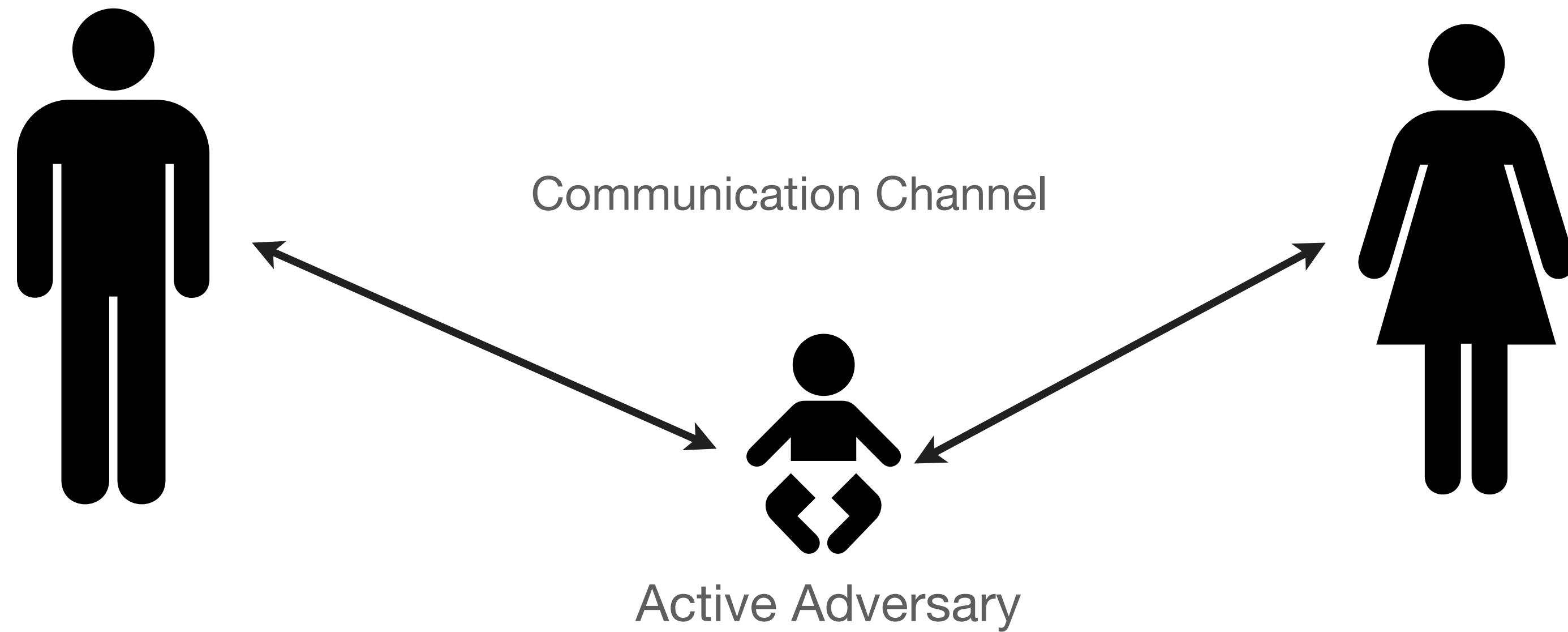
Communication Model



Communication Model



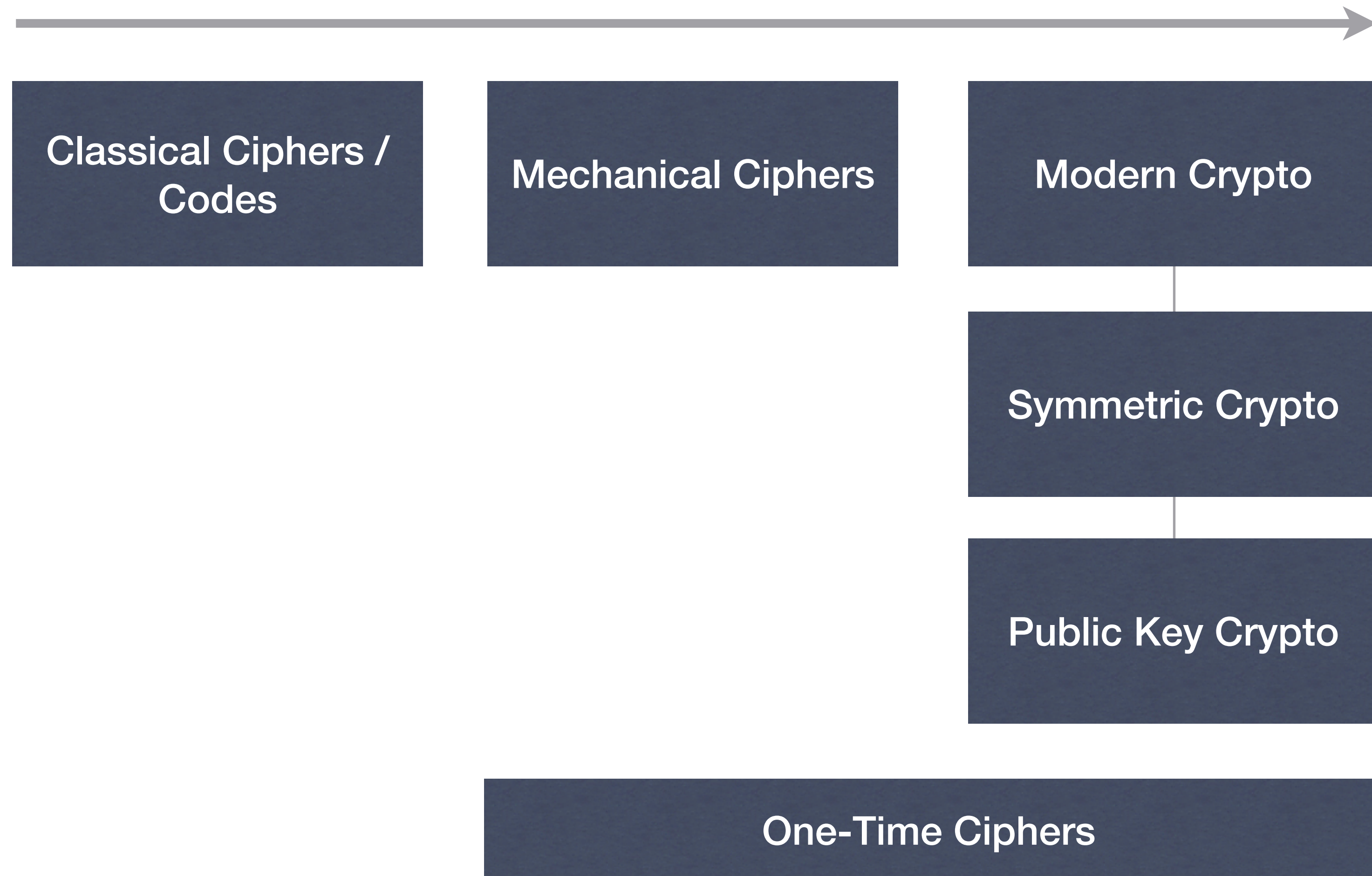
Communication Model



Secure Communication

- Two basic properties we like to achieve:
 - Data confidentiality
 - Data authenticity (“integrity”)
- Tools:
 - Encryption
 - Message Authentication Codes (MACs)
 - Digital Signatures

History of Encryption

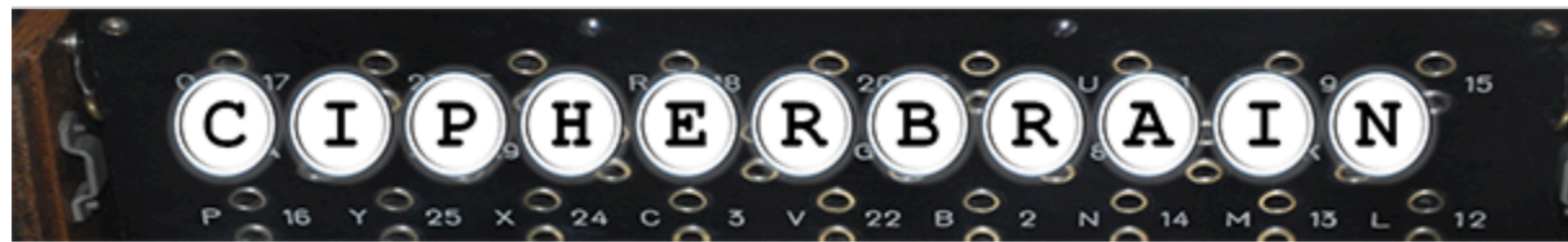


Classical Cryptography

- Beginning of time to 1900s or so
 - Shift (Caesar) cipher
 - Substitution ciphers
 - Polyalphabetic ciphers (Vigenère)
 - Digraph ciphers (Playfair)
 - A multitude of others...

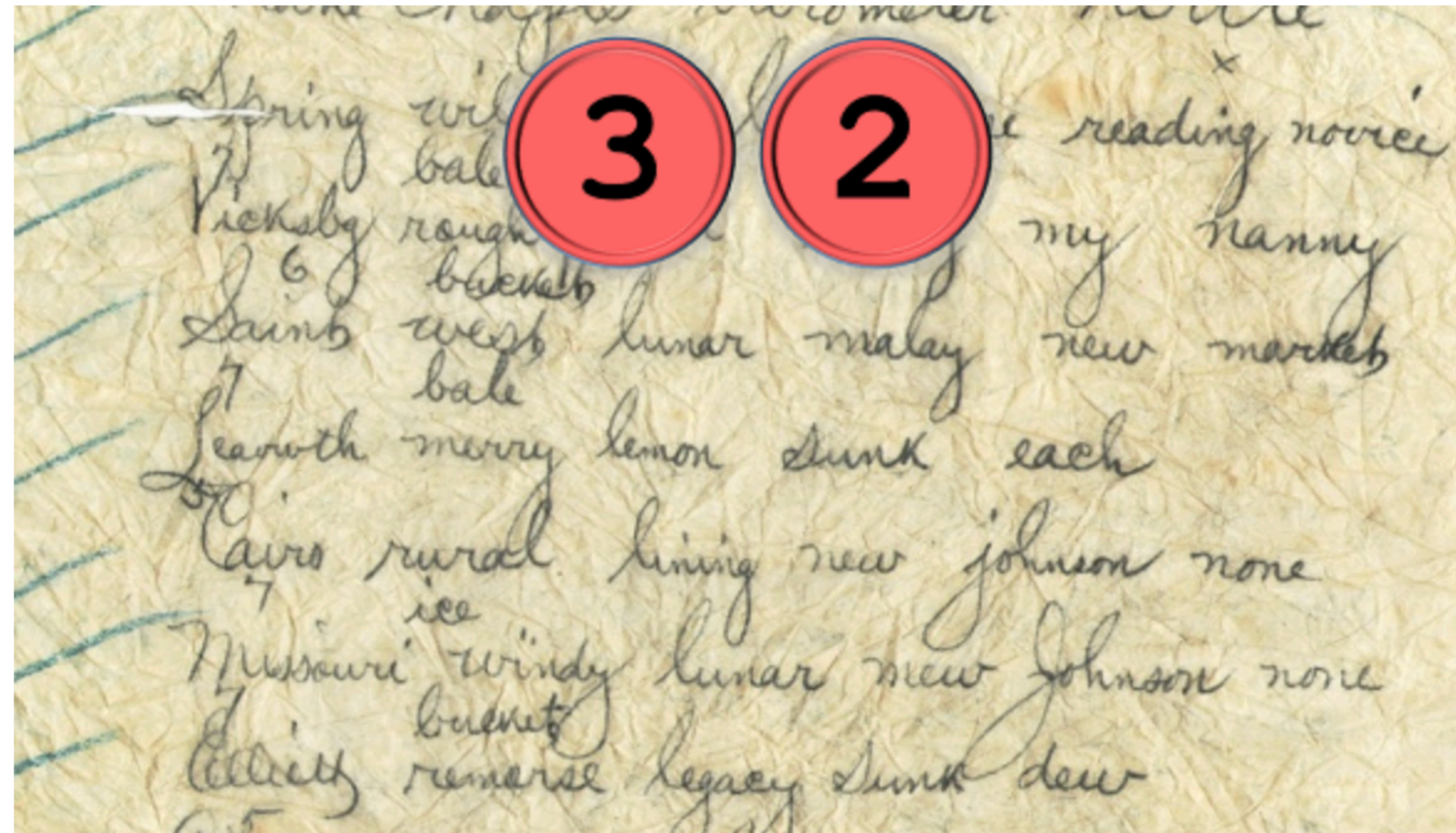


Increasing
Complexity



The Top 50 unsolved encrypted messages: 32. The silk dress cryptogram

Von [Klaus Schmeh](#) / 13. Mai 2017 / [6 Kommentare](#) / Seite 1 von 2 / [Auf einer Seite lesen](#)



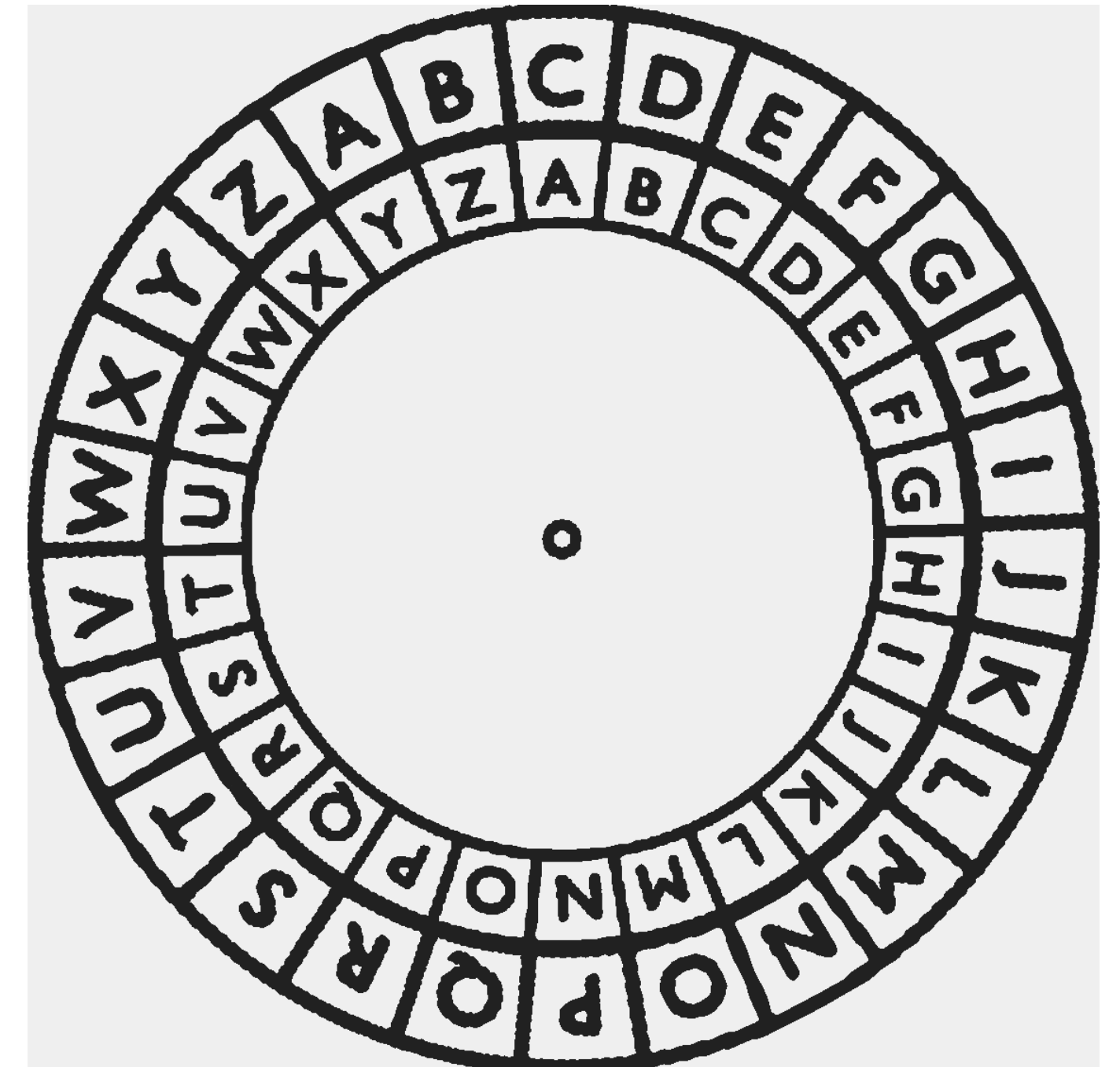
Shift (“Caesar”) cipher

- Key is a “shift value” (0 through 25)
- We take a plaintext and “shift” each letter through the alphabet:

Example, key = 10.

A T T A C K

K D D K M U



Shift (“Caesar”) cipher

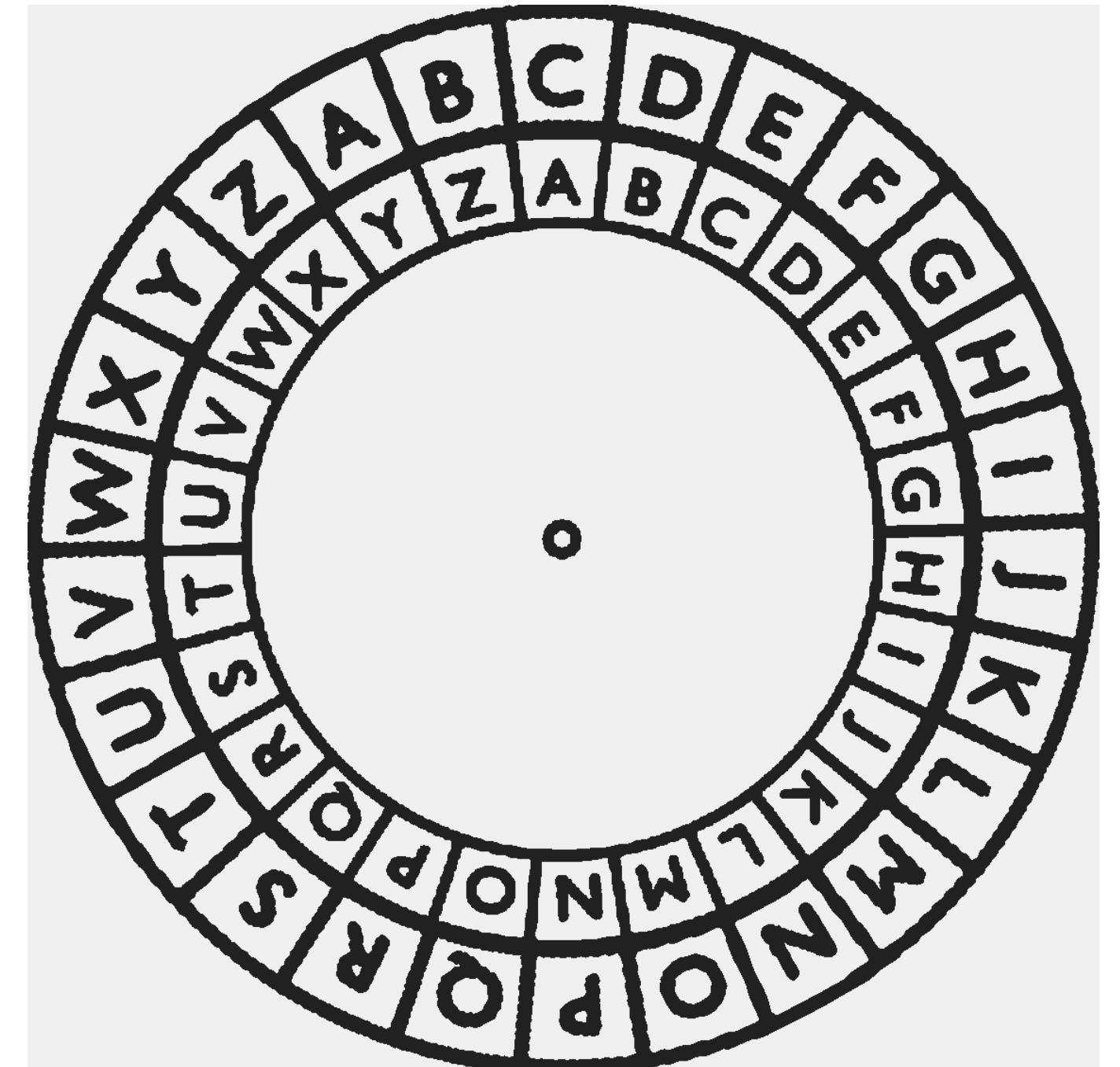
- Key is a “shift value” (0 through 25)
- We take a plaintext and “shift” each letter through the alphabet:

Example, key = 10.

A T T A C K

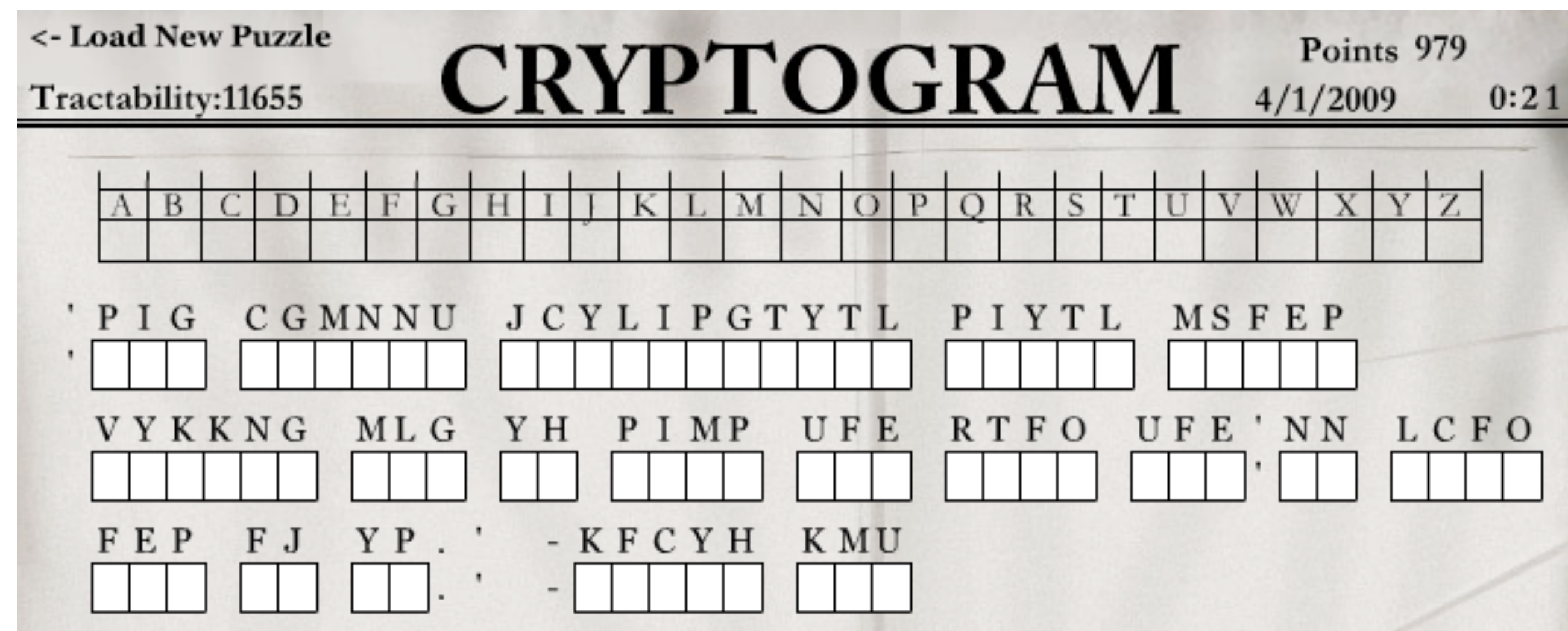
K D D K M U

Limitations of the shift cipher?



Substitution cipher

- Shift cipher has a small key space
 - This means it is vulnerable to “brute force attack”
- *We need a cipher with a larger set of possible keys*
- One solution is the substitution cipher



How many keys in an (alphabetic) substitution cipher?

How many keys in an (alphabetic) substitution cipher?

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
O	J	Z	E	R	K	U	A	S	B	P	W	L	H	M	X	V	Q	G	F	I	D	Y	C	T	N

Figure 2.1: Example key for a simple substitution cipher.

How many keys in an (alphabetic) substitution cipher?

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
O	J	Z	E	R	K	U	A	S	B	P	W	L	H	M	X	V	Q	G	F	I	D	Y	C	T	N

Figure 2.1: Example key for a simple substitution cipher.

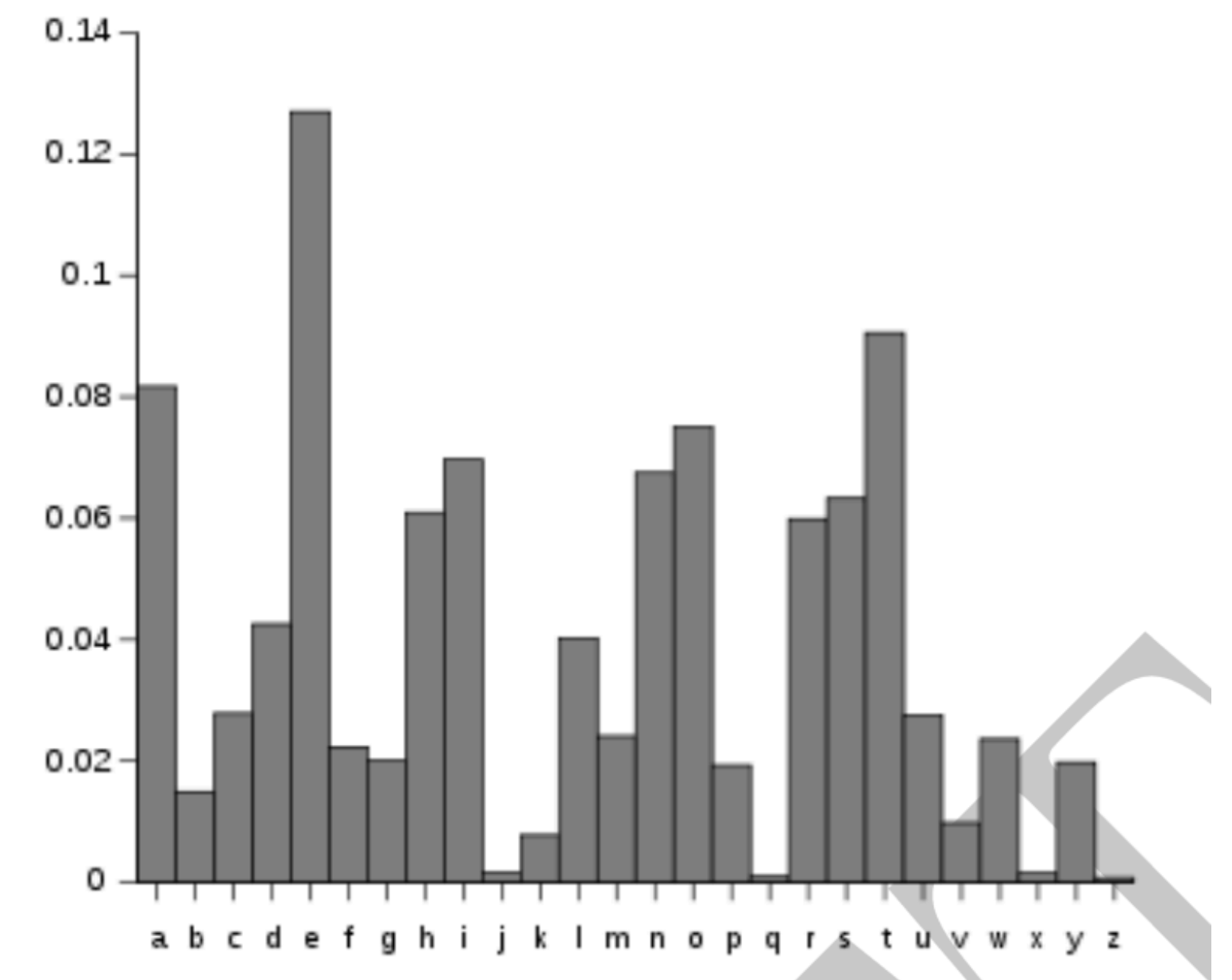
$$\text{\#keys} = 26 * 25 * 24 * \dots * 2 * 1$$

$$= 26! = 403 \text{ trillion trillion} = \text{about } 2^{88}$$

So what's wrong with substitution ciphers?

So what's wrong with substitution ciphers?

- Many things:
 - They leak letter patterns (e.g., repeated letters)
 - They leak letter frequency
- This is because the substitution “key” is the same for every letter



Vigenere

S	E	N	D	R	E	I	N	F	O	R	C	E	M	E	N	T	S
V	I	G	E	N	E	R	E	V	I	G	E	N	E	R	E	V	I
N	M	T	H	E	I	Z	R	A	W	X	G	R	Q	V	R	O	A

One-Time Ciphers

- 1900s
 - Vernam & Mauborgne's "Unbreakable" cipher
- Based on Baudot code for Teletypes
- Added (XORed) a random Key (sequence of bits) to a binary message
 - Perfectly secure, provided:
- key is perfectly random
- key is at least as long as the message
- key is never re-used



(No Model.)

11 Sheets—Sheet 6.

J. M. E. BAUDOT.
PRINTING TELEGRAPH.

No. 388,244.

Patented Aug. 21, 1888.

Fig. 24.

	1	2	3	4	5
A	+	—	—	—	—
B	—	—	+	+	—
C	+	—	+	+	—
D	+	+	+	+	—
E	—	+	—	—	—
F	+	+	—	—	—
G	—	+	+	+	—
H	—	+	—	+	—
I	+	+	—	+	—
J	—	+	+	—	—
K	+	—	—	+	—
L	+	—	—	+	+
M	+	+	—	+	+
N	—	+	+	+	+
O	+	+	+	—	—
P	+	+	+	+	+
Q	+	—	+	+	+
R	—	—	+	+	+
S	—	—	+	+	+
T	+	—	+	—	+
U	+	—	+	—	—
V	+	+	+	—	+
W	—	+	+	—	+
X	—	+	—	—	+
Y	—	+	—	—	+
Z	+	+	—	—	+
ε	—	—	—	+	—
ν	—	—	—	—	+

INVENTOR:

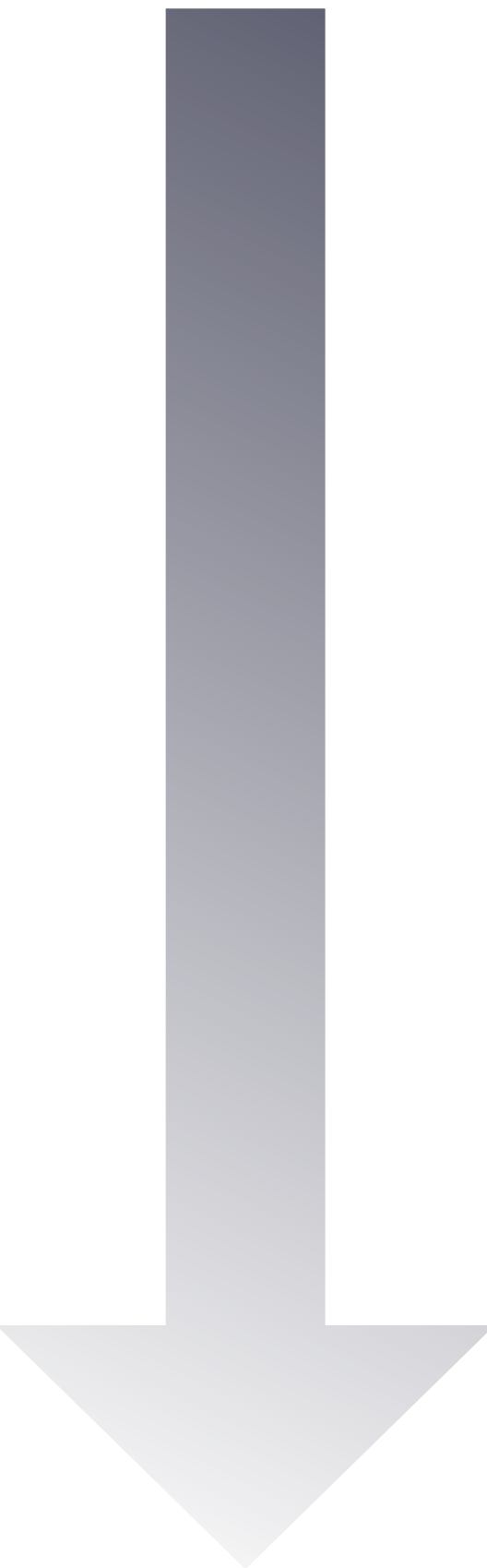
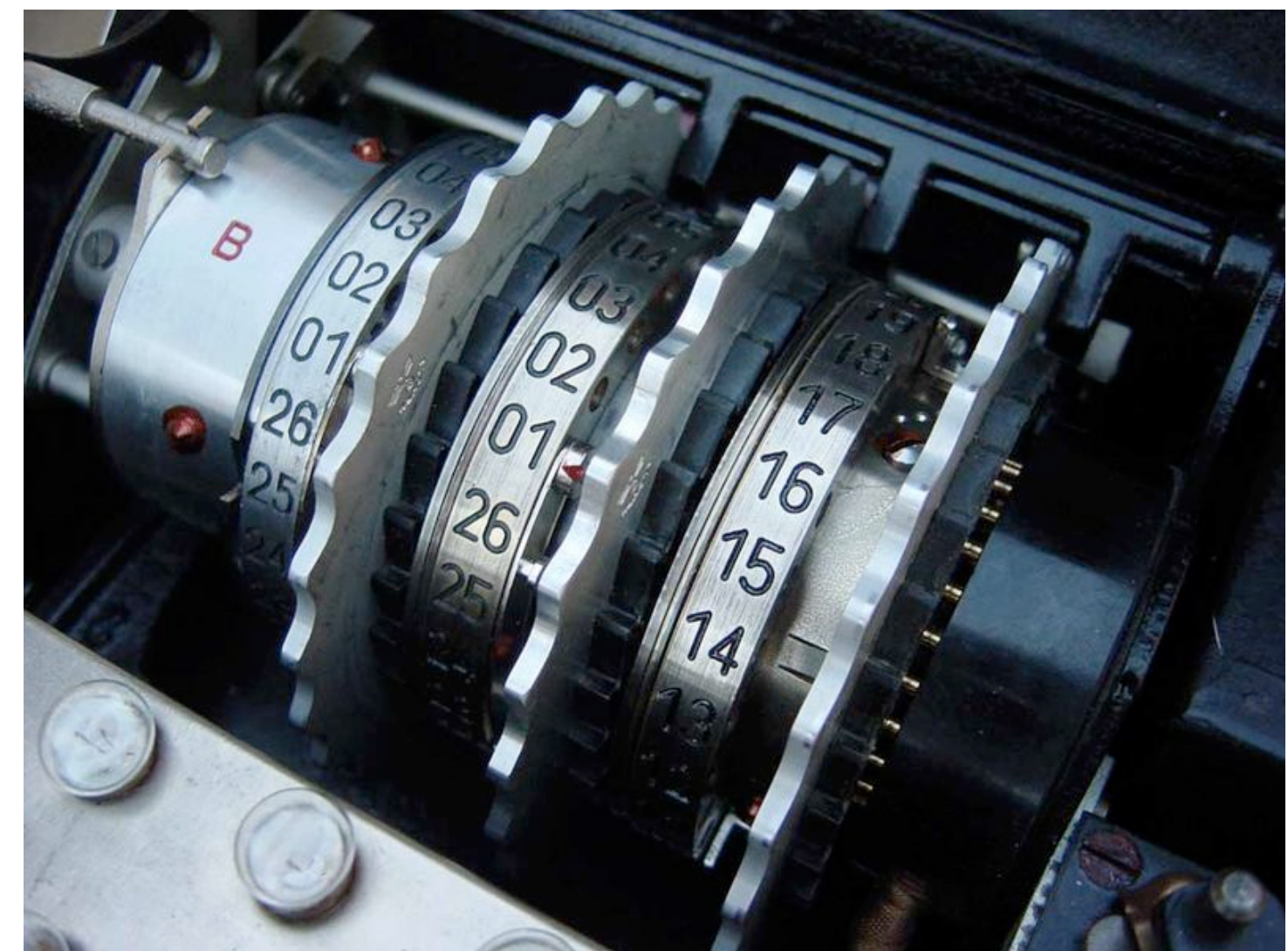
Jean Maurice Emile Baudot,

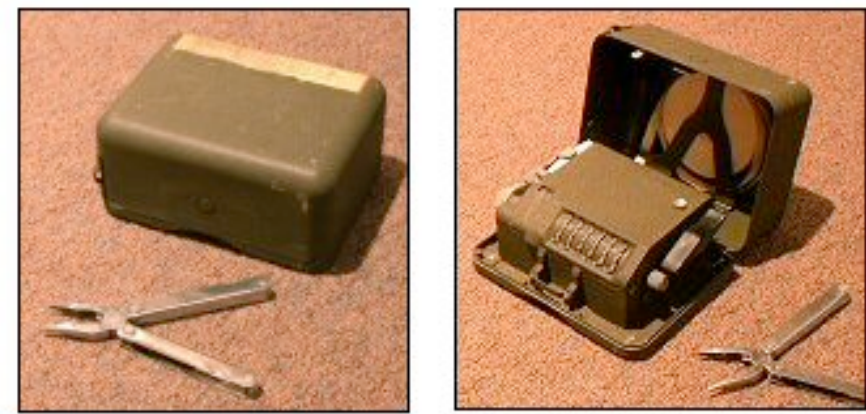
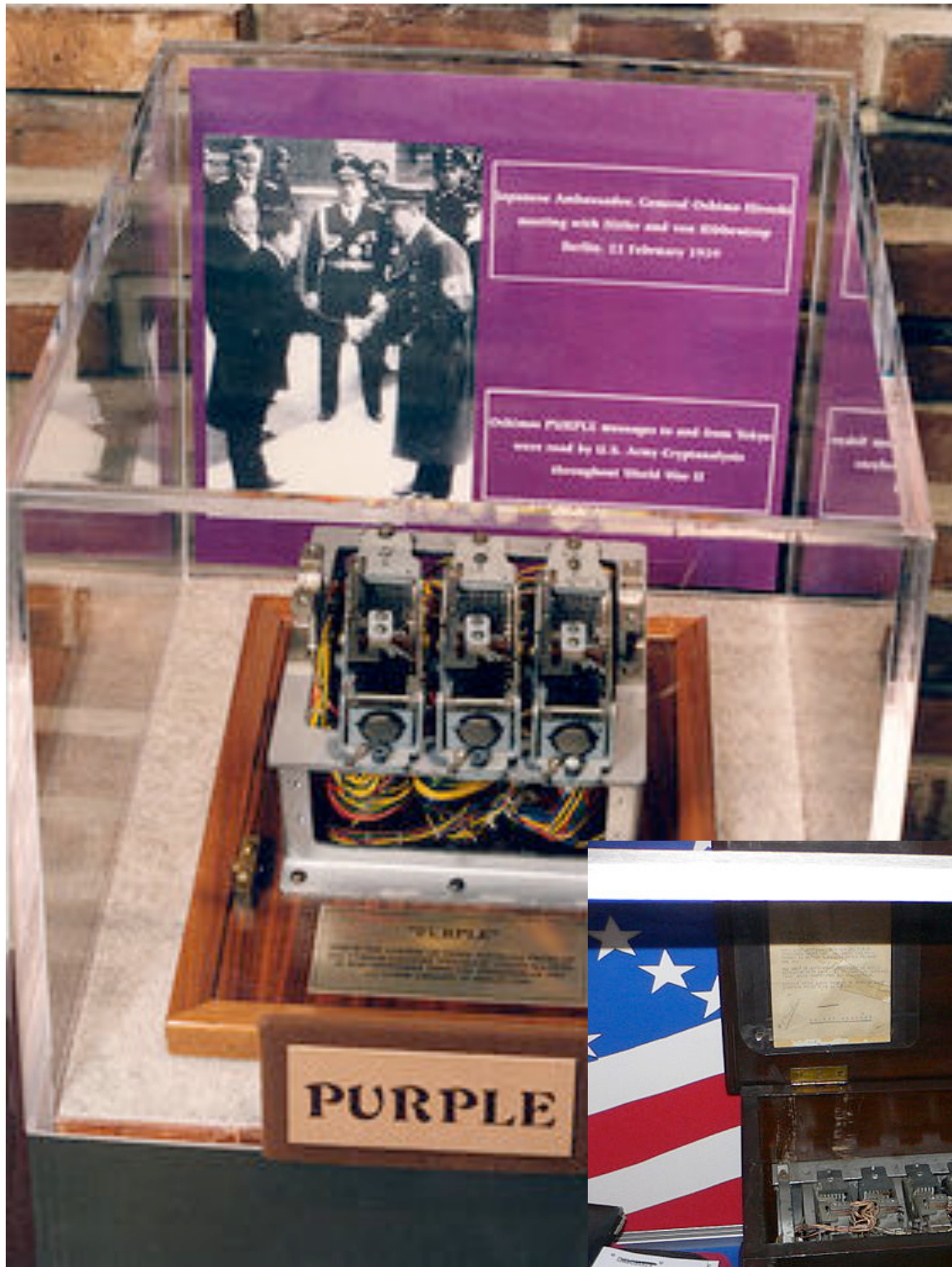




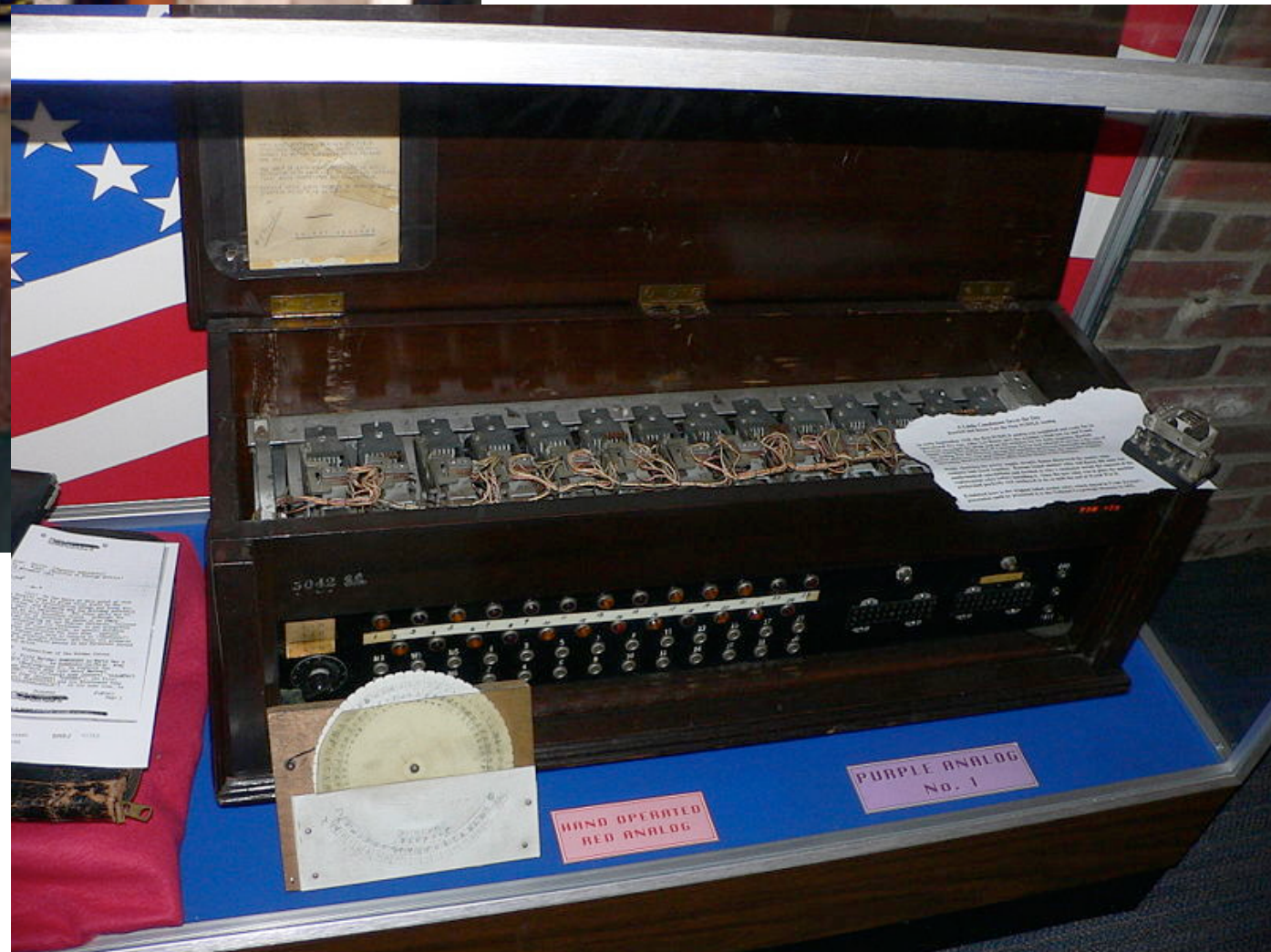
Mechanical Cryptography

- 1900s
 - Mass production and usage of cipher devices
 - Rotor ciphers
 - Electronic devices





HAGELIN M-209 CIPHER MACHINE (GVG / PD)





IYWJ2HOCX7PPDSE222OPXZYYXB1XFYSTT